Quantification: 876_z89c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	10
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2175.11	2169	869.225	px^2
major_axis_length_px	lacuna shape	84.7631	81.6741	19.9666	px
minor_axis_length_px	lacuna shape	33.7022	35.8518	7.7268	px
aspect_ratio	lacuna shape	2.557	2.7987	0.4604	unitless
eccentricity	lacuna shape	0.9107	0.934	0.04	unitless
solidity	lacuna shape	0.8171	0.8169	0.0544	unitless
roots_count	headline	11.6	12	5.7388	count
ring_length_r30_px	headline	674.6	716	152.425	px
ring_length_r60_px	reported	1755	1867	453.359	px
owned_length_px	ownership-dependent	6515.98	6637.34	2413.66	px
edge_count	ownership-dependent	481.5	479.5	178.927	count
mean_edge_length_px	ownership-dependent	13.5387	13.5865	0.4658	px
perimeter_px	normalised	274.683	274.09	105.205	px
ring_area_r30_px2	normalised	8917.1	8947	2293.42	px^2
ring_area_r60_px2	normalised	22387.9	23194	4941.33	px^2
in_frame_fraction_r30	normalised	0.9845	1	0.0489	unitless
in_frame_fraction_r60	normalised	0.958	0.9972	0.0818	unitless
ring_density_r30	normalised	0.0776069	0.0790836	0.0153553	px^-1
ring_density_r60	normalised	0.0792991	0.0807686	0.0163661	px^-1
roots_per_100px_perimeter	normalised	4.1308	3.9555	1.0953	per 100 px
ring_attached_length_r30_px	network v2	616.5	692.5	181.934	px
ring_attached_length_r60_px	network v2	1570.2	1685.5	589.04	px
ring_length_w_r30_px	network v2	761.157	815.535	173.3	px
ring_length_w_r60_px	network v2	1986.46	2126.45	522.445	px
sholl_crossings_r10	network v2	15.1	16	4.1486	count
sholl_crossings_r20	network v2	17.9	19	4.8865	count
sholl_crossings_r30	network v2	20.5	22	5.9114	count
loop_count	loops, report only	78.3	68.5	35.994	count
loop_edge_share	loops, report only	0.5647	0.5645	0.0918	share
ring_width_mean_r30_px	width	5.8503	5.6772	0.7901	px

Per field
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measure	value	units
analysed_area_px2	1.02878e+06	px^2
total_skeleton_length_px	73706	px
canalicular_length_density_per_px	0.0716444	px^-1
junction_count	2705	count
junction_density_per_px2	0.00262934	px^-2
lacuna_count	10	count

lacunae_per_px2	9.72e-06	px^-2
skeleton_component_count	542	count
mean_component_length_px	135.989	px
median_component_length_px	10	px
field_length_density_w_per_px	0.080685	px^-1
field_density_without_flagged_per_px	0.0751728	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	90	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0716444	px^-1
field_length_density_with_unattached_per_px	0.0717318	px^-1
loop_count	899	count
loop_edge_share	0.5297	share
edge_count	5773	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.262	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	61789	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3313	115.919	39.5634	2.93	0.94	0.7245	-1.5598	154.062	372.366	False
2	2754	105.554	35.8518	2.9442	0.9406	0.7558	1.5255	165.057	102.925	False
3	3301	102.585	44.3824	2.3114	0.9016	0.8169	-1.4051	307.062	909.902	False
4	2169	78.4025	38.343	2.0448	0.8723	0.8413	1.557	311.008	321.631	False
5	956	61.2609	21.1134	2.9015	0.9387	0.8	1.5273	461.978	61.1475	False
6	2610	87.8807	38.6915	2.2713	0.8979	0.8782	-1.531	535.001	198.764	False
7	1500	59.3512	33.3189	1.7813	0.8276	0.8966	1.5421	545.937	440.713	False
8	1670	81.6741	26.9572	3.0298	0.944	0.8337	-1.4297	692.65	612.224	False
9	1303	70.2403	25.0978	2.7987	0.934	0.8068	-1.3129	914.738	39.4635	False
10								71.8884	647.04	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	23	870	2363	9821.5	709	13.8526	5.2883
2	15	778	1852	4252.91	321	13.2489	5.6859
3	16	516	1268	7683.44	552	13.9193	5.0035
4	12	807	2098	5321.55	399	13.3372	6.7128
5	12	724	1641	3740.62	290	12.8987	6.9903
6	13	753	2188	7507.19	575	13.056	6.7179
7	6	708	2031	9832.18	753	13.0573	6.3834
8	9	697	1882	8038.6	581	13.8358	5.6684
9	7	487	1107	3194.32	228	14.0102	4.818

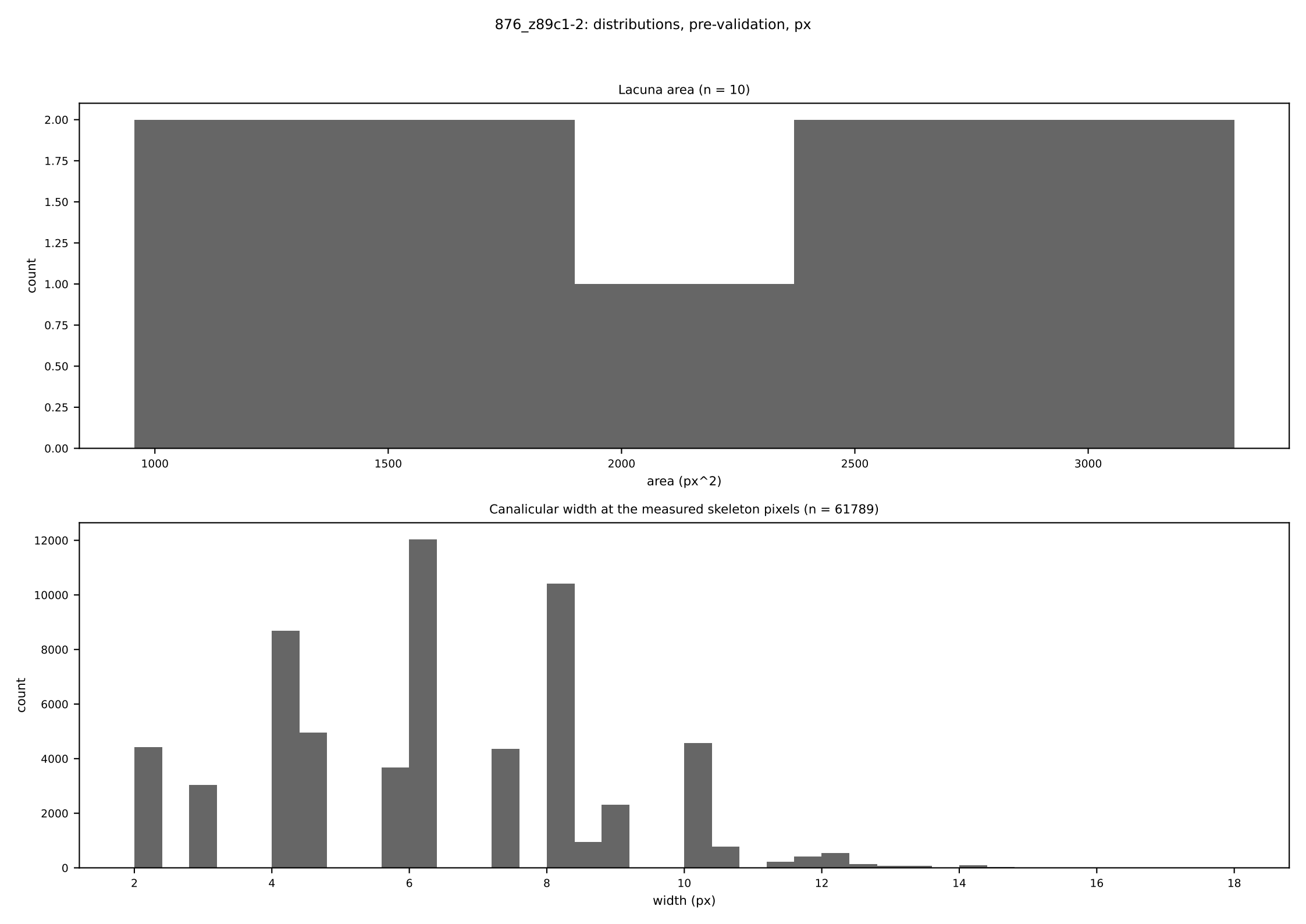
10 3 406 1120 5767.49 407 14.1707 5.2343

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	423.014	12381	29774	1	1	0.070269	0.0793646	5.4372
2	413.328	11320	26755	1	0.9628	0.0687279	0.0692207	3.6291
3	373.665	11365	27711	1	0.9875	0.0454025	0.045758	4.2819
4	273.505	9180	23731	1	1	0.0879085	0.0884076	4.3875
5	191.315	7221	16849	1	0.887	0.100263	0.0973945	6.2724
6	277.848	9459	23485	1	1	0.0796067	0.0931658	4.6788
7	190.593	7507	20585	1	1	0.094312	0.0986641	3.1481
8	274.676	8714	22903	1	1	0.0799862	0.0821726	3.2766
9	236.043	6856	16232	0.8455	0.748	0.0710327	0.0681986	2.9656
10	92.8467	5168	15854	1	0.9944	0.0785604	0.0706446	3.2311

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	797	2280	966.482	2647.2	19	22	28
2	720	1660	874.826	2104.34	21	20	22
3	411	1035	584.387	1431.66	11	15	18
4	781	1922	915.683	2378.7	18	20	22
5	685	1550	824.671	1845	18	18	22
6	731	2182	861.754	2505.51	15	20	26
7	700	2008	806.399	2323.93	17	17	22
8	659	1711	783.671	2148.56	13	26	24
9	284	580	539.647	1238.39	10	11	11
10	397	774	454.049	1241.34	9	10	10


876_z89c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 61789)
count
width (px)